Type 1: Research Project

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Motion Consistency Evaluation in Unmodified Surveillance Video

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1. Feedback Integration (Panel Review 1)

Panel Feedback Received

- **No specified use-case** — the problem lacked a concrete, motivated deployment scenario
- **Methodology too simple** — sliding window statistical model is insufficient; panel expects at least a deep learning model
- **Scene ambiguity** — in an open parking lot, a stationary car could be mistaken for a frozen feed; no mechanism to differentiate

Our Response

- Revised use case to **Traffic** CCTV networks — large-scale, safety-critical, zero-tolerance for feed failure
- Replaced the handcrafted C_t formula with an **Adaptive Dual-Stream LSTM Autoencoder** — two independent streams (frame differencing and optical flow) with a novel divergence anomaly score $\text{Score} = e_A + e_B + \lambda\,|e_A - e_B|$; threshold self-calibrates per camera via a dual-window rolling baseline.
- Addressed scene ambiguity using **pre-pipeline guard layers** — pixel variance (G_1) and SSIM inter-frame similarity (G_2) catch genuine freeze before the model runs.

2. Problem Statement

Problem Context

Surveillance camera systems generate continuous video streams, but real-world feeds often suffer from hidden issues such as prolonged low motion, frozen frames, or irregular temporal behavior due to environmental or hardware problems. These issues may not trigger traditional object-based alerts yet significantly reduce monitoring reliability.

This project evaluates motion consistency in unaltered surveillance videos by analyzing natural temporal motion patterns to detect irregularities and assess camera feed health and reliability.

3. Demonstration of Gaps

What Existing Systems Do

- Focus on **activity recognition** — detecting crimes, intrusions, or events
- Assume the camera feed is always **healthy and reliable**
- Use object detection or human annotation — **computationally expensive**

What is Missing

- No dedicated system to monitor **camera feed integrity itself**
- Statistical models use **manual thresholds** — not scalable across cameras
- No unified approach to classify **fault type** (freeze vs duplicate vs irregular)
- No **learned reliability score** — existing flags are binary (healthy/faulty)
- Cannot improve with more data — **no learning capability**
- No system uses **dual-stream divergence** to cross-validate feed health

*Existing literature monitors **what happens in front of the camera** — not **whether the camera itself is working**.*

4. Demonstration of Need

Real World Problem

- Urban traffic camera networks operate **500+ feeds** simultaneously across intersections
- Manual monitoring of each feed is **practically impossible** at city scale
- A frozen or duplicated feed gives a **false sense of security** during live incidents
- Faulty feeds during critical events lead to **zero forensic evidence**
- Camera faults go **undetected for hours** in large deployments

Why Automation is Critical

- Human operators experience **attention fatigue** within minutes
- Network-wide faults (e.g. server freeze) affect **multiple cameras at once**
- Real-time alerting requires **sub-second fault detection**
- Maintenance teams need **precise fault type** — not just a binary alert
- Reliability score enables **prioritized response** across the camera network

An automated, learned, real-time camera health monitoring system is not a luxury — it is a ***critical infrastructure requirement****.*

5. Challenges

Technical Challenges

- **Freeze vs low-motion scenes** — empty corridors resemble frozen feeds; pixel variance and SSIM guards required
- **Gradual degradation** — slow feed decline absorbed by rolling baseline; dual-window threshold mitigates
- **Dynamic lighting** — day/night cycles affect motion features; grayscale normalization applied
- **No labeled fault datasets** — ground truth for camera faults is scarce; synthetic fault injection used
- λ **selection** — divergence weight has no analytic solution; grid search over $\{0.1, 0.5, 1.0, 2.0\}$

System-Level Challenges

- **Manual threshold setting** — per-camera calibration not scalable; rolling adaptive threshold addresses this
- **Optical flow computation cost** — Farneback on full resolution is slow; frames resized to 64×64 before flow
- **Camera heterogeneity** — different resolutions, frame rates, placements require warm-up calibration period
- **Class imbalance** — healthy feeds vastly outnumber faulty ones; synthetic injection balances training data
- **Privacy restrictions** — real surveillance data is difficult to obtain; public dataset (AI City challenge) used

6. Literature Survey – Part 1

Related Works (Papers 1–4)

Paper Title	Author/Year	Problem Focus	Essence
Inter-Frame Tampering Detection via Texture & Optical Flow	Akhtar et al., 2024	Inter-frame tampering detection	Proposes a two-stage pipeline combining LBP-based texture analysis with optical flow consistency checks to detect and localize frame-level tampering in surveillance videos.
Blind Video Quality Assessment based on Spatio-Temporal Feature Resolver (STFR-BVQA)	Bi et al., 2024	Blind video quality assessment	Introduces a universal spatio-temporal feature resolver using ZCA and ICA, followed by SVR to predict video quality without relying on deep learning models.
Deep Learning-Based Detection and Localization of Inter-Frame Video Tampering	Huu-Thanh Duong et al., 2023	Inter-frame inconsistency detection	Employs deep spatio-temporal feature representations to detect and localize frame insertion, deletion, and duplication attacks in surveillance footage.
Frame Duplication Forgery Detection in Surveillance Videos	Li et al., 2023	Frame duplication detection	Uses wavelet-domain analysis combined with texture features to identify and localize duplicated frames in surveillance video sequences.

6. Literature Survey – Part 2

Related Works (Papers 5–9)

Paper Title	Author/Year	Problem Focus	Essence
LSTM-Based Video Quality Prediction for Temporal Distortions	Mittag et al., 2023	Quality prediction with frame freezes/skips.	LSTM models temporal distortions to predict MOS.
Frame Shuffling Forgery Detection Method for MPEG-Coded Video	Kharat J. P. et al., 2024	Detection of frame shuffling forgery in MPEG videos.	Uses motion vector discontinuity to detect suspicious frames and validates them using SIFT feature matching.
Vision Transformer Embedded VAD	Ummay Maria et al., 2025	Deep anomaly detection using transformers	Learns long-range spatial and temporal motion dependencies using transformer-based attention mechanisms.
AnomLite: Efficient Video Anomaly Detection	Anna K. Zvereva et al.,2025	Lightweight video anomaly detection	Efficient spatio-temporal motion modeling with reduced computational complexity.
Video Quality Assessment with Texture Information Fusion	Menon et al., 2024	Efficient VMAF estimation for streaming.	Uses DCT-based texture features and SSIM fused with an LSTM model to estimate VMAF faster and more efficiently.

7. Research Direction

Existing Approaches

- Surveillance systems primarily focus on detecting **abnormal events or behaviors**.
- Temporal inconsistencies are commonly treated as **scene-level anomalies**.
- Video forensic methods target **intentional frame tampering**.
- Video quality techniques address **network or compression impairments**.
- The **camera feed itself is generally assumed reliable**.

Proposed Research Direction

- Shift from **scene-centric** to **camera-centric** reliability analysis.
- Analyze **natural temporal motion patterns** in surveillance videos.
- Detect silent camera issues such as **freezing, low motion, or instability**.
- No reliance on **object detection or anomaly labels**.
- Applicable to **unmodified real-world surveillance feeds**.

8. Originality - novel idea

- We propose a **camera health assessment method** based on temporal motion pattern analysis in surveillance video streams.
- Unlike traditional approaches focused on object or event detection, our framework enables:
  - Continuous surveillance feed monitoring
  - Non-intrusive evaluation
  - No object detection requirement
  - No human annotation dependency
- The system evaluates **real-world surveillance feeds** purely through motion consistency modeling.

9. Objectives & Deliverables

Objectives

- Automatically detect when a surveillance camera feed is **frozen, degraded, or irregular** without human intervention
- Distinguish between a **genuinely still scene** and a **broken camera feed**
- Learn normal camera behaviour from **unlabeled footage** and self-calibrate per camera — no manual setup required
- Assign each feed a continuous **reliability score** and classify it as Healthy, Frozen, or Irregular

Deliverables

- **D1:** Literature survey and gap analysis report
- **D2:** Preprocessed dataset with synthetically injected fault annotations
- **D3:** Trained dual-stream model with guard layers and divergence scoring
- **D4:** Evaluation report with Precision, Recall, F1, Reliability Score, and ablation study

10. Proposed Methodology – Core Concept

Adaptive Dual-Stream LSTM-AE with Divergence Anomaly Scoring

Why upgrade from statistical model?

- Previous C_t formula is a **z-score** – brittle
- Manual threshold τ set
- Cannot distinguish **scene stillness vs freeze**
- Single stream misses **partial faults**
- No learning capability

New approach

- **Guard layers** catch freeze before AE runs
- **Dual LSTM-AE** on two independent streams
- **Divergence score** cross-validates both streams
- **Dual-window threshold** adapts per camera
- **LSTM** models temporal motion patterns

Output: Feed Health + Reliability Score

10. Proposed Methodology – Processing Pipeline (1/2)

Guard Layer G1 – Pixel Variance

- Compute pixel variance across $T = 16$ frames
- $\mathrm{Var} < \theta_v \Rightarrow$ **FROZEN** (skip AE)

Guard Layer G2 – SSIM Check

- Compute $\mathrm{SSIM}(F_t, F_{t-1})$ per frame pair
- $\mathrm{SSIM} > 0.999$ for ≥ 8 frames $\Rightarrow$ **FROZEN**

Stage 1 – Preprocessing

- Resize to 64×64 · grayscale · normalise $[0, 1]$
- Frame diff: $M_t^A = \|F_t - F_{t-1}\|$
- Farneback optical flow magnitude: M_t^B
- Pre-computed offline and cached

Stage 2 – Clip Builder

- Sliding window $T = 16$ frames
- Stream A: $\{M_1^A, \ldots, M_{16}^A\}$
- Stream B: $\{M_1^B, \ldots, M_{16}^B\}$

Stage 3 – Dual LSTM-AE

- Two independent autoencoders
- Encoder: $16 \rightarrow 64$ units; Decoder: $64 \rightarrow 16$
- Trained on normal footage only – no labels needed
- Outputs: reconstruction errors e_A, e_B

10. Proposed Methodology – Processing Pipeline (2/2)

Stage 4 – Divergence Anomaly Score

- $\text{Score} = e_A + e_B + \lambda\,|e_A - e_B|$
- λ via grid search: $\{0.1,\ 0.5,\ 1.0,\ 2.0\}$
- Novel contribution – divergence term catches faults single-stream misses

Stage 5 – Dual-Window Rolling Threshold

- Short window ($N_s = 20$): detects sudden faults
- Long window ($N_l = 100$): detects gradual drift
- $\theta = \min(\mu_s + 2\sigma_s,\ \ \mu_l + 2\sigma_l)$
- Self-calibrates per camera – no manual tuning

Stage 6 – Classification

- G1 or G2 triggered $\Rightarrow$ **Frozen**
- $\text{Score} > \theta \Rightarrow$ **Irregular**
- Otherwise $\Rightarrow$ **Healthy**
- Feed Reliability Score $R \in [0, 1]$

*Trained on **normal footage only** – no fault labels required.*
Synthetic fault injection used for evaluation only.

10. Proposed Methodology – Mathematical Model (1/2)

Guard Layer G1: Pixel Variance

$$\mathrm{Var}_i = \frac{1}{T}\sum_{t=1}^{T}(F_t - \bar{F})^2$$

FROZEN if $\mathrm{Var}_i < \theta_v$

Guard Layer G2: SSIM Similarity

$\mathrm{SSIM}(F_t,\ F_{t-1}) > 0.999$ for $\geq$8 frames $\Rightarrow$ FROZEN

Stage 1: Motion Streams

$$M_t^A = \|F_t - F_{t-1}\|, \qquad M_t^B = \|\mathrm{flow}(F_t,\ F_{t-1})\|$$

Stage 3: Dual LSTM-AE

$$e_A = \|\mathrm{s}^A - \mathrm{Dec}_A(\mathrm{Enc}_A(\mathrm{s}^A))\|^2$$

$$e_B = \|\mathrm{s}^B - \mathrm{Dec}_B(\mathrm{Enc}_B(\mathrm{s}^B))\|^2$$

LSTM Temporal Modelling

$$h_n = \mathrm{LSTM}(\mathrm{v}_n,\ h_{n-1})$$

$$s_n = \sigma(W_o h_n + b_o)$$

Previous model (replaced):

$$C_t = \frac{|M_t - \mu_{\mathrm{window}}|}{\sigma_{\mathrm{window}} + \epsilon}$$

with fixed threshold τ

10. Proposed Methodology – Mathematical Model (2/2)

Stage 4: Divergence Anomaly Score (Novel)

$$\text{Score}_i = e_A + e_B + \lambda\,|e_A - e_B|$$

$$\lambda \in \{0.1,\ 0.5,\ 1.0,\ 2.0\} \quad \text{(grid search)}$$

Stage 5: Dual-Window Adaptive Threshold

$$\theta_s = \mu_s + 2\sigma_s \quad (N_s = 20)$$

$$\theta_l = \mu_l + 2\sigma_l \quad (N_l = 100)$$

$$\theta = \min(\theta_s,\ \theta_l)$$

Feed Reliability Score

$$R = \frac{1}{N}\sum_{n=1}^{N} s_n \ \in [0,1]$$

Stage 6: Classification Rule

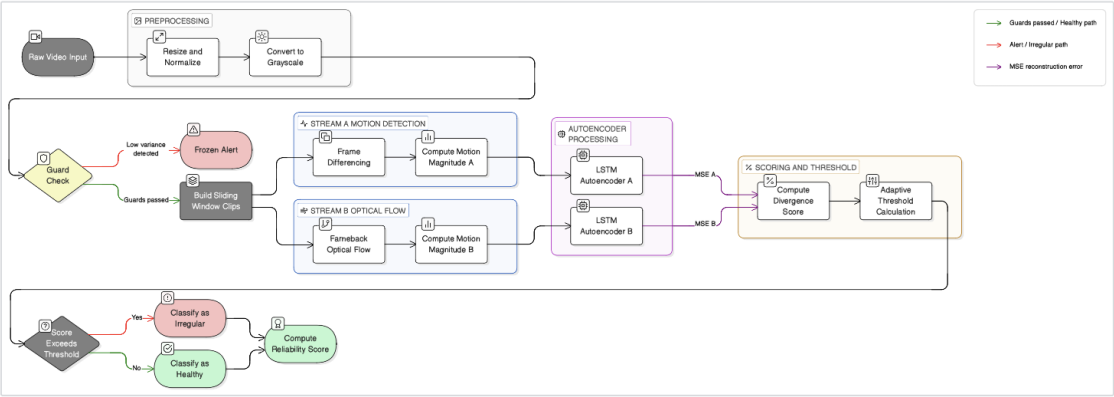
$$\text{Class}_i = \begin{cases} \textbf{Frozen} & \text{if G1 or G2 triggered} \\ \textbf{Irregular} & \text{if } \text{Score}_i > \theta \\ \textbf{Healthy} & \text{otherwise} \end{cases}$$

Stage 6: Classification Loss

$$\mathcal{L} = -\sum_k y_k \log \hat{y}_k$$

$k \in \{\text{Healthy},\ \text{Frozen},\ \text{Irregular}\}$

11. Conceptual Architecture – Flow Diagram



12. Dataset Overview

Dataset Used in This Work – AI City Challenge Traffic Anomaly Dataset

Property	Details
Full Name	AI City Challenge – Traffic Anomaly Detection Dataset
Source	NVIDIA AI City Challenge (CVPR Workshop, annual)
Type	Real-world urban traffic surveillance video
Size	100 training videos + 150 test videos; each ≈15 min long; 62+ hours total
Camera Setup	Multiple fixed traffic cameras; intersections, stretches of roadway, and highways
Resolution	800×410 (standard); up to 1080p in extended tracks
Frame Rate	30 fps
Anomaly Types	Vehicle crashes (single & multi-vehicle), stalled vehicles
Annotation	Temporal ground truth – anomaly start & end timestamps per video
Key Properties	Fixed-camera traffic surveillance; long normal segments; multi-camera city-scale deployment
Usage in Our Work	Healthy-feed motion baseline training for Dual LSTM-AE;
Access	`https://www.aicitychallenge.org/ai-city-challenge-dataset-access/`

13. Dataset Sample Images

Sample Images from Key Dataset

FV33 Daytime highway traffic scene with moderate vehicle flow under clear conditions

FV051 Low-light/nighttime traffic scene with reduced visibility and active vehicle movement

FV 072 Urban highway segment showing steady traffic flow

14. Distribution of Work

PHASE 1 — Current

Research & Definition
Traffic CCTV · Gap · Architecture

KEY TASKS
- Problem statement & use-case
- Literature survey & gap analysis
- Dual LSTM-AE methodology design
- Architecture diagram & PPT
- Dataset: ShanghaiTech + AI City

DELIVERABLES
D1: Survey & Gap Report
D2: Architecture Document
D3: Dataset Preparation Plan

PHASE 2 — Next semester

Implementation
& Development

KEY TASKS
- Preprocessing pipeline (64×64)
- Guard layers G1 + G2
- Dual LSTM-AE training
- Divergence anomaly scoring
- Fault injection (freeze, dup, deg)

DELIVERABLES
D4: Guard Layer Module
D5: Dual LSTM-AE Module
D6: Divergence Scoring Module

PHASE 3 — Final semester

Validation
& Reporting

KEY TASKS
- Evaluation on injected test set
- End-to-end classification
- Precision, Recall, F1, Score R
- Ablation: with / without λ
- Final report

DELIVERABLES
D7: Full Integrated Pipeline
D8: Evaluation Report
D9: Final Research Report

15. Conclusion & Future Work

Future Work & Enhancements

- **Cross-camera consistency verification** — extend the framework to compare motion embeddings across spatially adjacent camera feeds; a hardware freeze contradicts its neighbours, a genuinely static scene does not.
- **3×3 spatial grid motion extraction** — divide each frame into a 3×3 grid and compute per-region motion scalars, enabling detection of partial frame corruption and localised faults invisible to global scalar features.
- **Meta-network adaptive thresholding** — replace the rolling $\mu + 2\sigma$ baseline with a small learned network that ingests a camera's motion profile (mean activity, variance, time-of-day patterns) and outputs a personalised threshold; handles cameras with strongly non-stationary baselines.
- **Edge-optimised deployment** — quantise and prune the dual LSTM-AE for real-time inference on edge IoT hardware (e.g. NVIDIA Jetson), enabling on-device feed health monitoring at metro / railway scale without central server dependency.

11. References

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3	Blind Video Quality Assessment on Spatio-Temporal Features *Bi et al.*	2024
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8	Multi-Camera Spatiotemporal DL for Behavior Detection *Veesam et al.*	2025
9	LSTM Video Quality Prediction for Temporal Distortions *Mittag et al.*	2023
10	Inter-Frame Tampering Detection via Texture & Optical Flow *Akhtar et al.*	2024
11	Future Frame Prediction for Anomaly Detection *Liu et al.*	2023
12	Anomaly Detection via Global-Local Information *Zhang et al.*	2023

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14	Multi-Sensor Surveillance with Secure Alerting *Hussain et al.*	2024
15	SurveillanceVQA-589k: Video-Language Benchmark *Liu et al.*	2025
16	Deep-STA: Inter-Frame Tampering Detection *Kaddah et al.*	2022
17	Robust Motion Detection Using Motion Vectors *Ali et al.*	2023
18	Blind VQA: Spatio-Temporal Resolver *Bi et al.*	2024
19	Video QA with Texture Information Fusion *Zhang et al.*	2024
20	Frame Shuffling Forgery Detection for MPEG *Wang et al.*	2023
21	Proactive Anomaly Surveillance for CCTV *Sarker et al.*	2024
22	SecureCam: Selective Detection & Encryption *Vora et al.*	2023

Thank You

Questions & Discussion